

Airframe Technology Master

Structural Test and Materials

Lecture 04

Introduction to metallic materials (II)

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13 June 2025

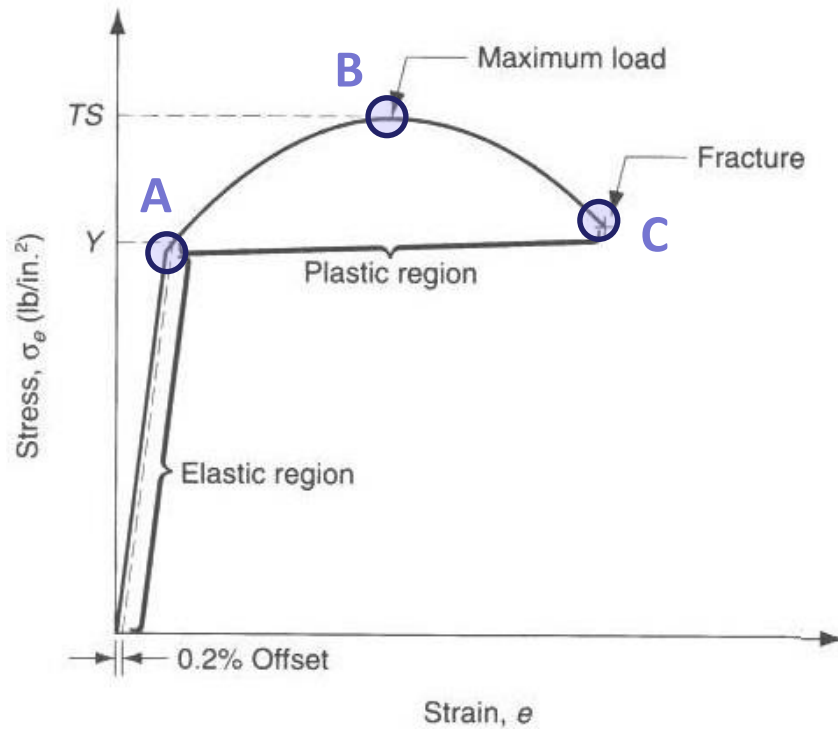
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Stress-Strain curve



Depends on:

Application

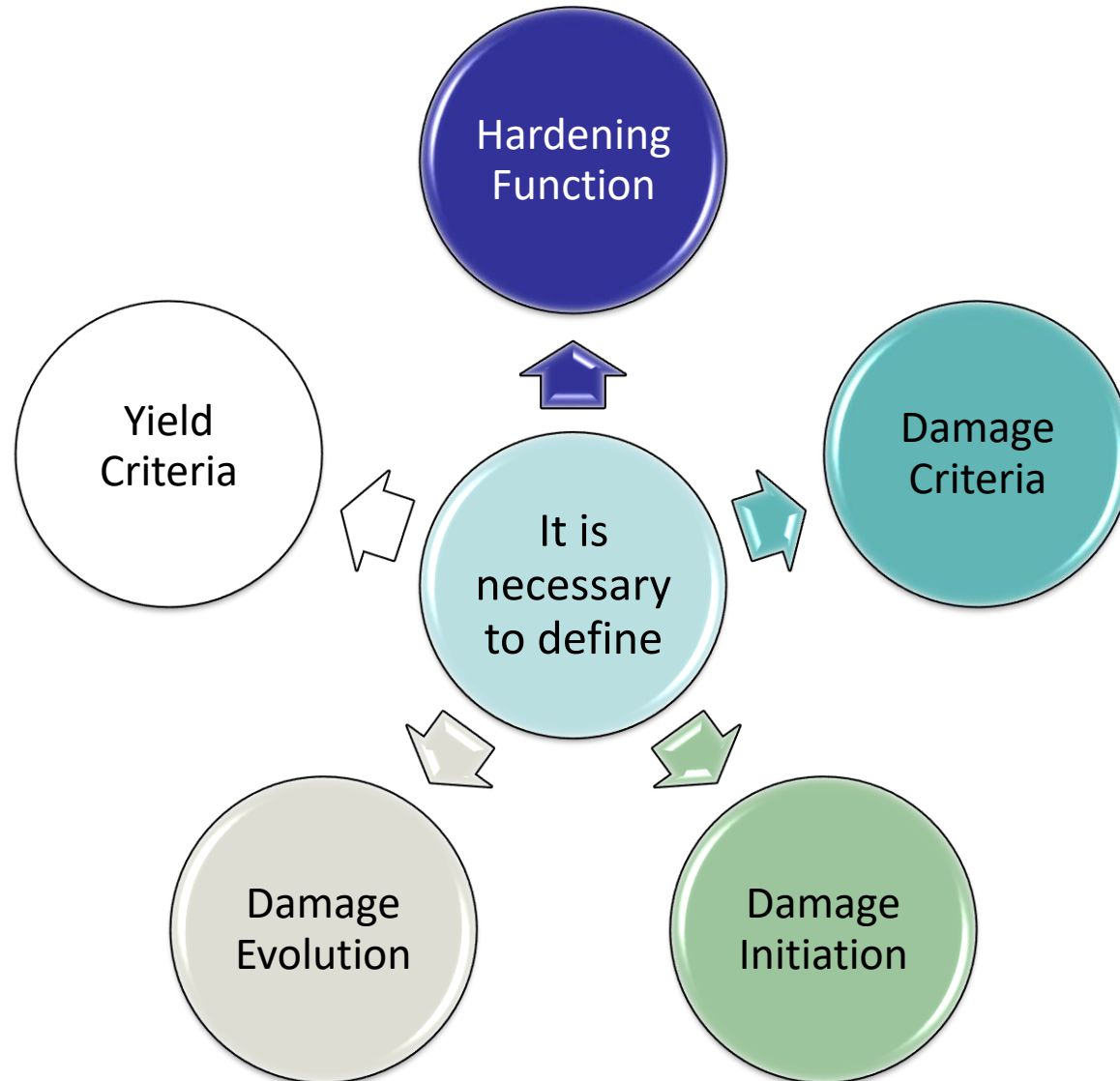
Material



Depends on:

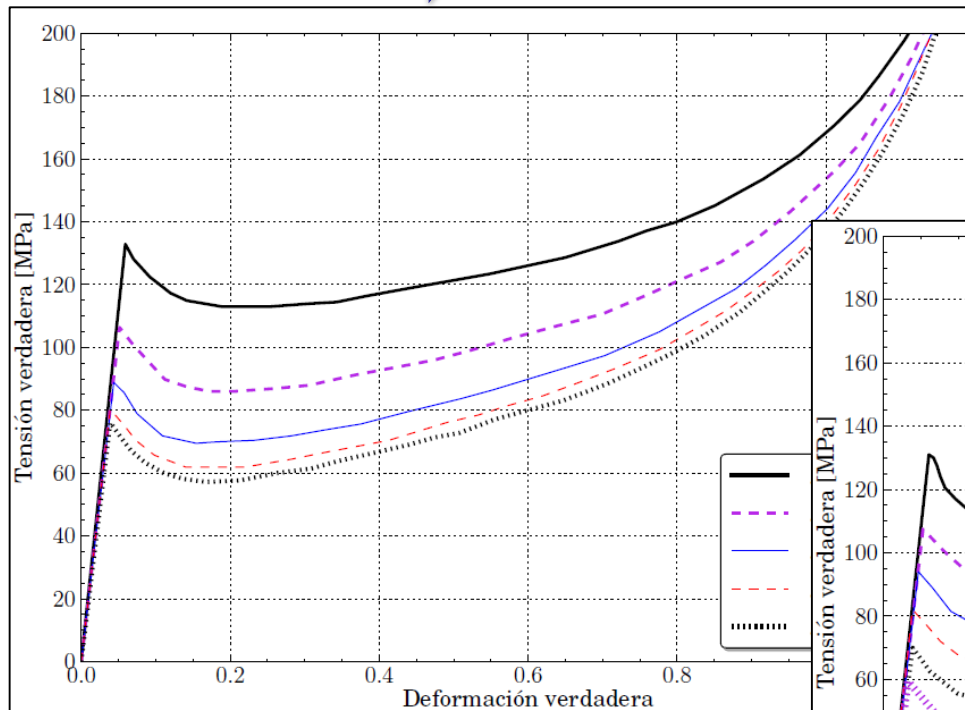
Application

Material

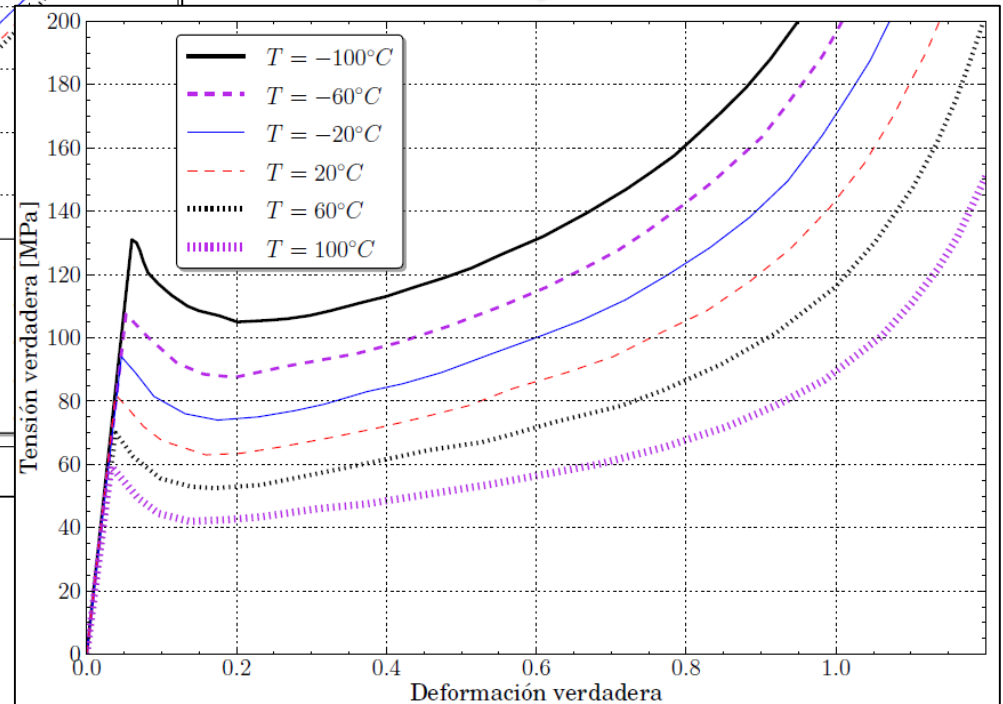


Take into account the constitutive equation

Strain rate  hardening



Temperature  softening

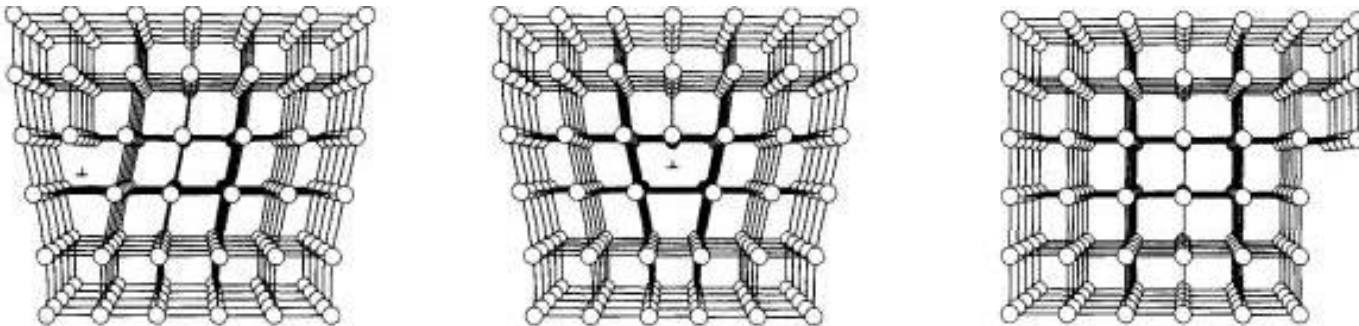


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Micromechanical approach

- Plasticity is characterized by the appearance of **permanent deformations** (or plastic deformations), which do not recover when the load ceases.
- These deformations correspond to **relative displacements between atoms**, displacements that are **stable** when the load is removed.



- The origin of plastic deformations is the presence and, particularly, the movement of **dislocations** in the crystalline lattice.
- The process by which plastic deformation is produced by the movement of dislocations is called "**sliding**".
- It is essentially associated with **shear forces** (in general, with **deviatoric stresses**).

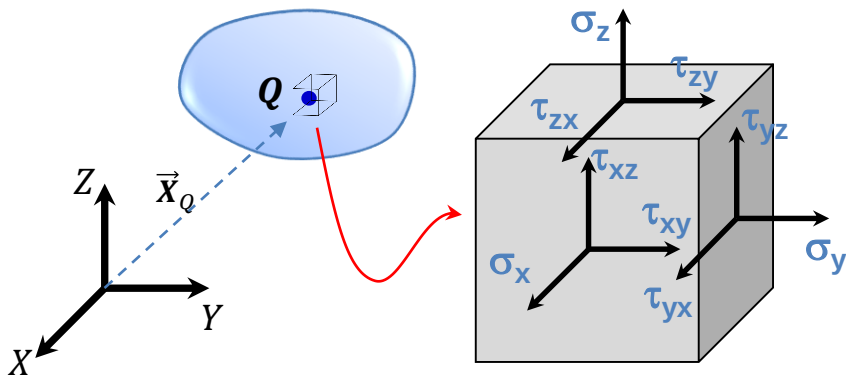
Yield Criteria

In a **uniaxial tensile test**, the yield criterion is:

$$\sigma - \sigma_Y = 0$$

If the stress state is **multi-axial**, we must generalize the concept of **yield limit**.

Be a solid subjected to a general (possibly multi-axial) stress state at a point **Q**:



Yield Function:

It is a scalar function that depends on the components of the stress tensor

$$f = f(\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{xz}, \tau_{yz})$$

Yield Criterion:

- In **elastic regime**, the yield function is **strictly negative**

$$f(\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{xz}, \tau_{yz}) < 0$$

- At the **limit of the elastic regime**, the yield function is **cancelled**.

$$f(\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{xz}, \tau_{yz}) = 0$$

Basic Assumptions

- The effect of **time** is not considered.
- The effects produced by the **lack of homogeneity** of the material on a microscopic scale are not considered: the **actual material** is idealized as a **homogeneous continuous medium**.

Additional Assumptions

- The material is **isotropic**.
- Plasticity is **independent of the hydrostatic** stress component.
- The **Bauschinger effect** is not considered



$$f(J_2) = 0$$

The stress state associated with Q can be represented by the vector $\overrightarrow{OQ}$, with components:

$$\overrightarrow{OQ} = (\sigma_1, \sigma_2, \sigma_3)^T$$

This vector can be decomposed into its components *perpendicular* and *parallel* to the plane π :

$$\overrightarrow{OQ} = \overrightarrow{OP} + \overrightarrow{PQ}$$

Where $\overrightarrow{OP}$ has the form:

$$\overrightarrow{OP} = (\sigma_p, \sigma_p, \sigma_p)^T$$

Establishing orthogonality between $\overrightarrow{PQ}$ and a vector $\vec{u}$ parallel to the diagonal:

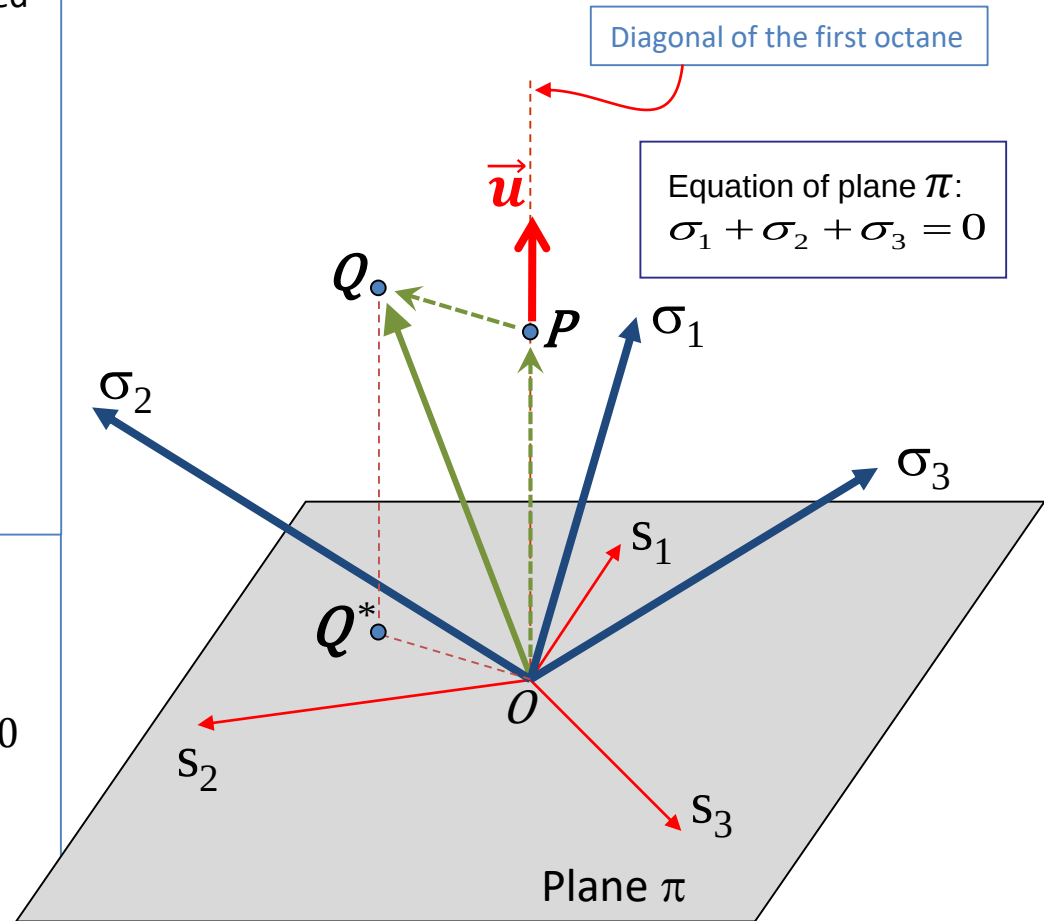
$$\overrightarrow{PQ} \cdot \vec{u} = (\sigma_1 - \sigma_p, \sigma_2 - \sigma_p, \sigma_3 - \sigma_p) \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix} = 0$$

It can be obtained:

$$\sigma_p = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3} = \frac{I_1}{3} = \bar{\sigma}$$

With the above, it can be obtained:

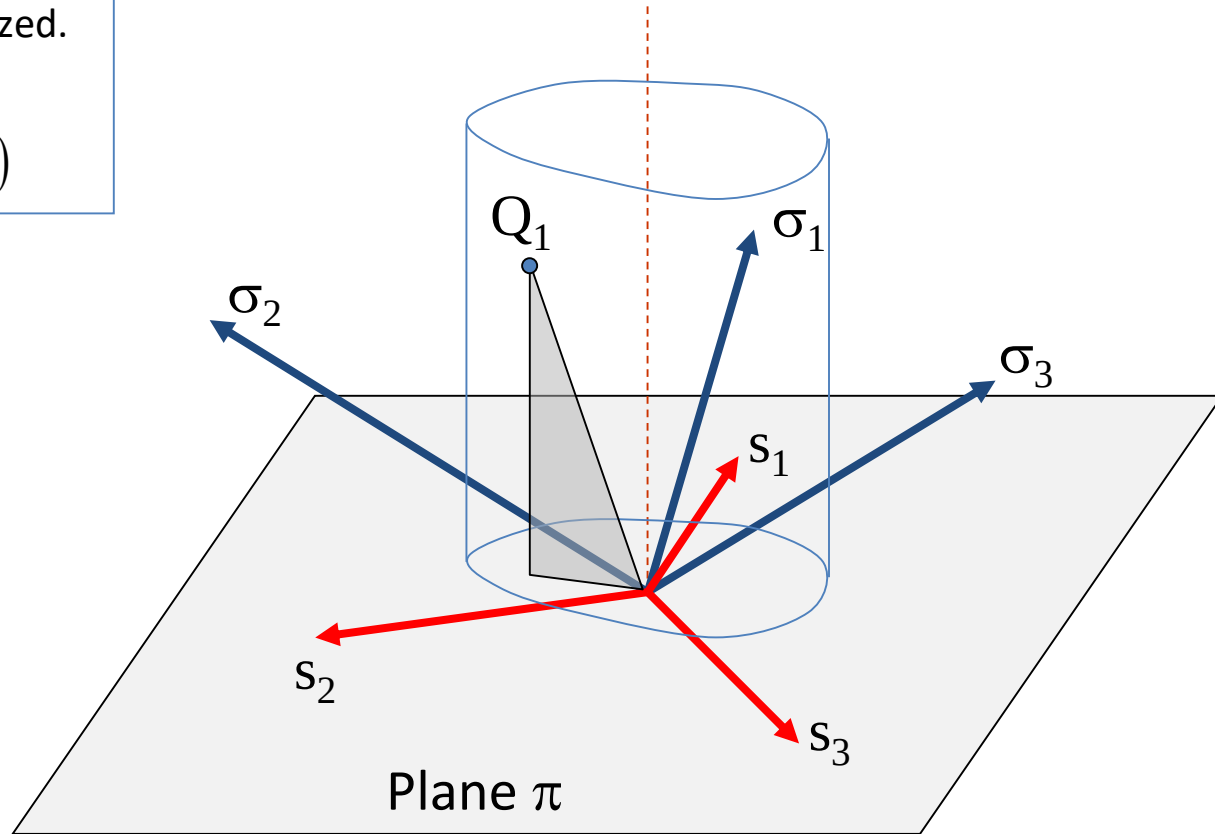
$$\overrightarrow{OP} = (\bar{\sigma}, \bar{\sigma}, \bar{\sigma})^T = \frac{1}{3} (I_1, I_1, I_1)^T \quad ; \quad \overrightarrow{PQ} = (\sigma_1 - \bar{\sigma}, \sigma_2 - \bar{\sigma}, \sigma_3 - \bar{\sigma})^T = (s_1, s_2, s_3)^T = \overrightarrow{OQ^*}$$



Yield Criteria

Q_1 : The material has not plasticized.
 Q_1 is inside the yield surface

$$f(J_2) < 0 \quad \text{for} \quad J_2 = J_2(Q_1)$$



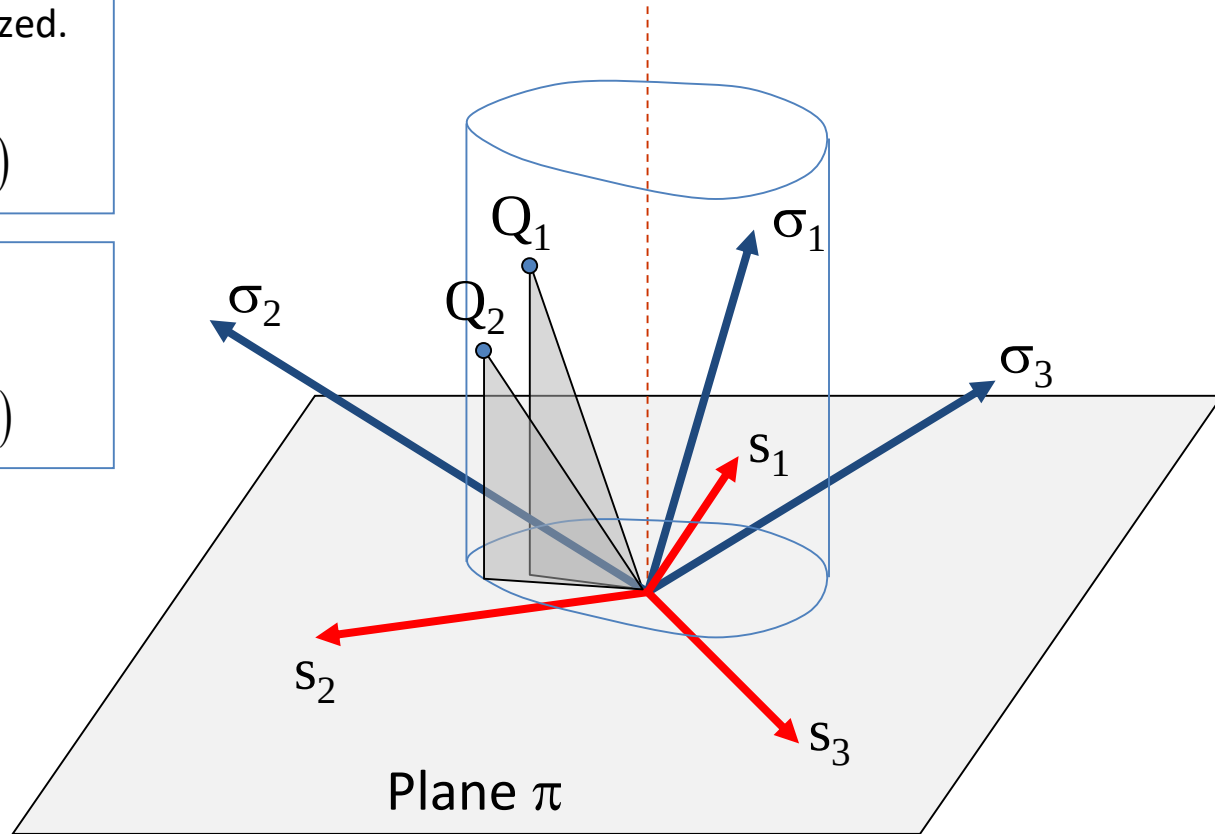
Yield Criteria

Q_1 : The material has not plasticized.
Q1 is inside the yield surface

$$f(J_2) < 0 \quad \text{for} \quad J_2 = J_2(Q_1)$$

Q_2 : The material has plasticized.
Q2 is inside on yield surface

$$f(J_2) = 0 \quad \text{for} \quad J_2 = J_2(Q_2)$$



Yield Criteria

Q_1 : The material has not plasticized.
 Q_1 is inside the yield surface

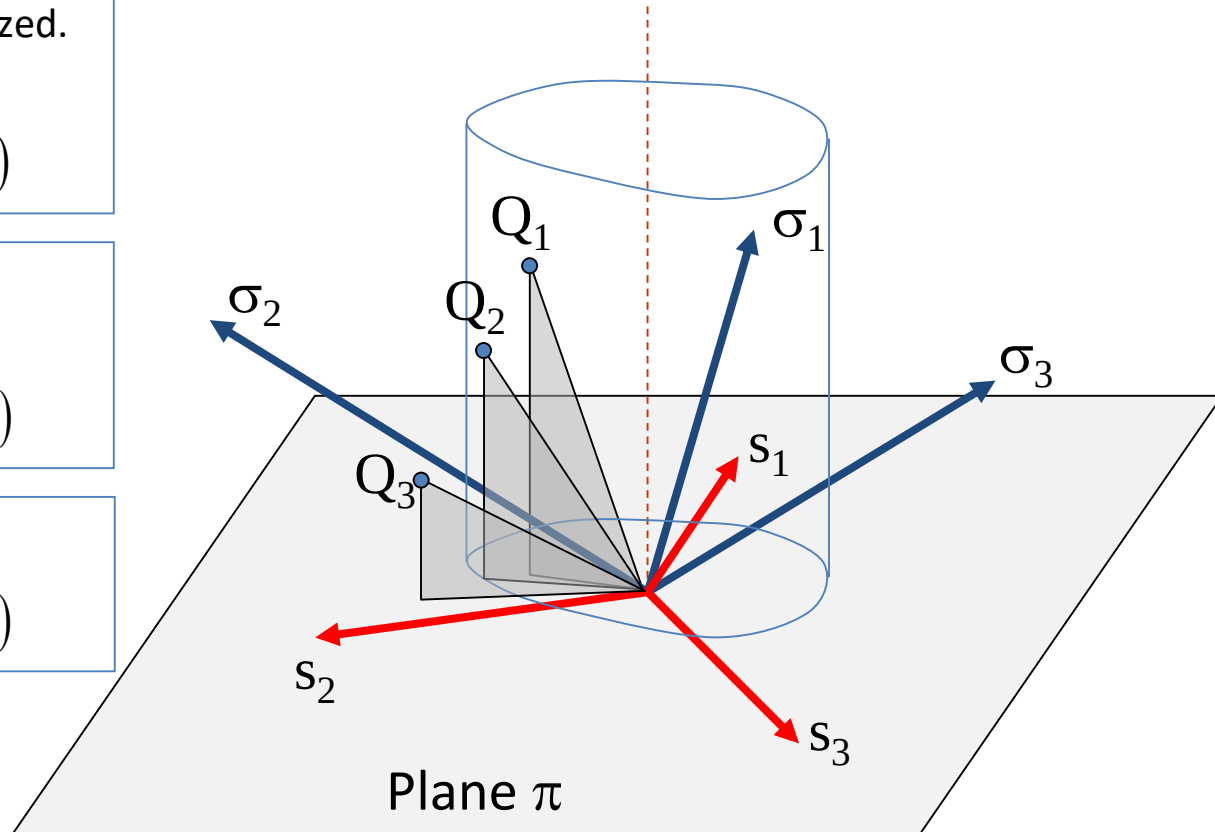
$$f(J_2) < 0 \quad \text{for} \quad J_2 = J_2(Q_1)$$

Q_2 : The material has plasticized.
 Q_2 is on the yield surface

$$f(J_2) = 0 \quad \text{for} \quad J_2 = J_2(Q_2)$$

Q_3 : Is it possible?.

$$f(J_2) > 0 \quad \text{for} \quad J_2 = J_2(Q_3)$$



Yield Criteria

Q_1 : The material has not plasticized.
 Q_1 is inside the yield surface

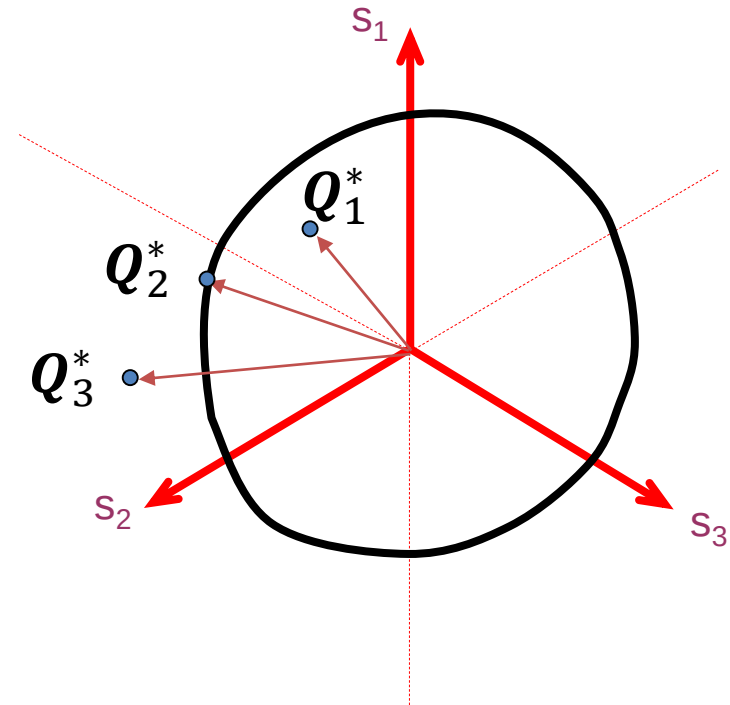
$$f(J_2) < 0 \quad \text{for} \quad J_2 = J_2(Q_1)$$

Q_2 : The material has plasticized.
 Q_2 is on the yield surface

$$f(J_2) = 0 \quad \text{for} \quad J_2 = J_2(Q_2)$$

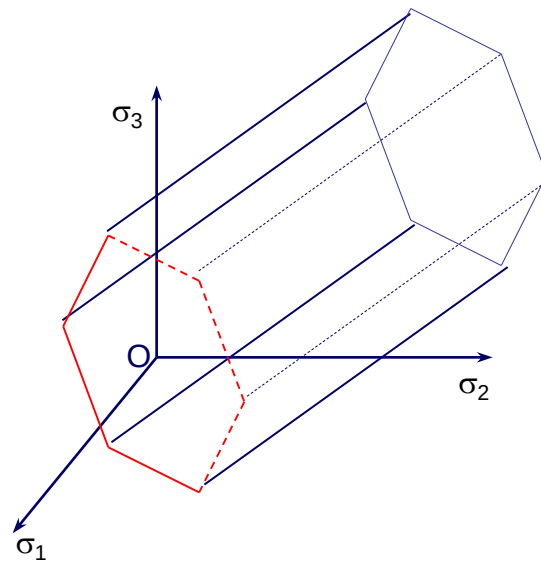
Q_3 : Is it possible?.

$$f(J_2) > 0 \quad \text{for} \quad J_2 = J_2(Q_3)$$

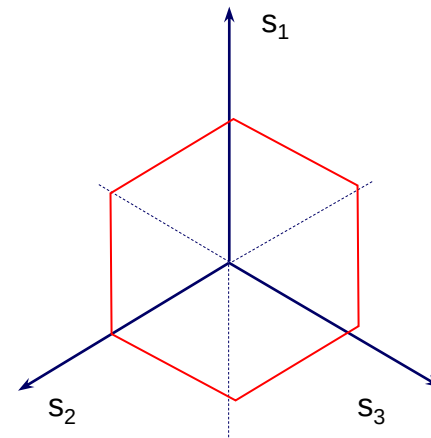


Tresca

$$\tau_{max} = (\sigma_1 - \sigma_3)/2 = \sigma_Y/2 \quad \text{being} \quad \sigma_1 \geq \sigma_2 \geq \sigma_3$$



Yield surface



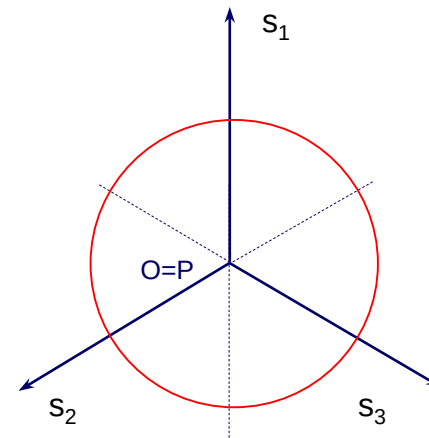
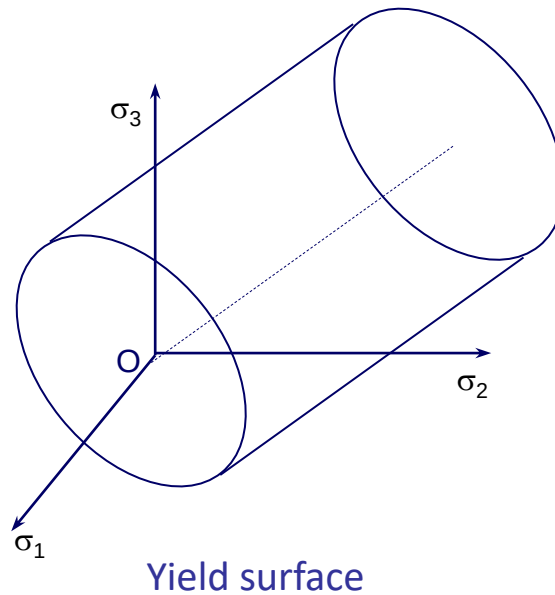
Cut between the yield surface
and the deviatoric plane

Yield locus

$$f(J_2) = (\sigma_1 - \sigma_3) - \sigma_Y$$

Von Mises

$$\sqrt{3J_2} = \sigma_Y$$



Cut between the yield surface
and the deviatoric plane

Yield locus

$$f(J_2) = \sqrt{3J_2} - \sigma_Y$$

Rankine-Lame

$$|\sigma_1| < \sigma_Y$$

$$|\sigma_2| < \sigma_Y$$

$$|\sigma_3| < \sigma_Y$$

Saint Venant-Poncelet

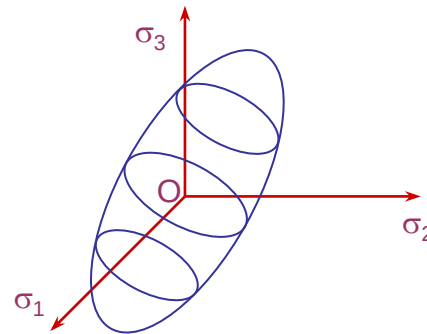
$$|\sigma_1 - \nu(\sigma_2 + \sigma_3)| < \sigma_Y$$

$$|\sigma_2 - \nu(\sigma_1 + \sigma_3)| < \sigma_Y$$

$$|\sigma_3 - \nu(\sigma_1 + \sigma_2)| < \sigma_Y$$

Beltrami-Haig

$$I_1^2 + 2 \cdot (1 + \nu) \cdot I_2 < \sigma_Y^2$$



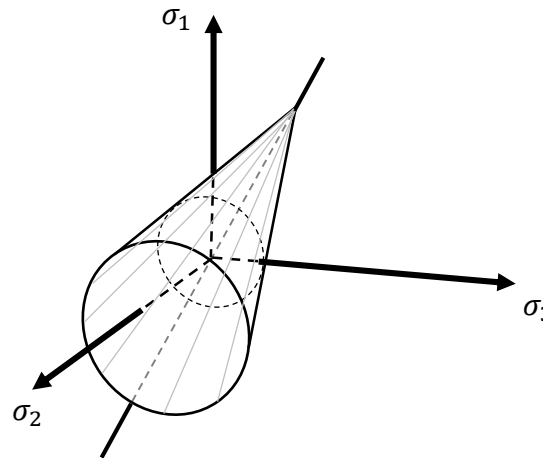
Drucker-Prager

$$\sqrt{3J_2} + \alpha I_1 - \beta = 0$$

It can be considered as a generalization of the Von Mises criterion, adding the linear dependence of the first invariant of the stress tensor.

The parameter α is dimensionless and corresponds to an **internal friction coefficient**.

The parameter β has stress units and corresponds to a **cohesion coefficient**.



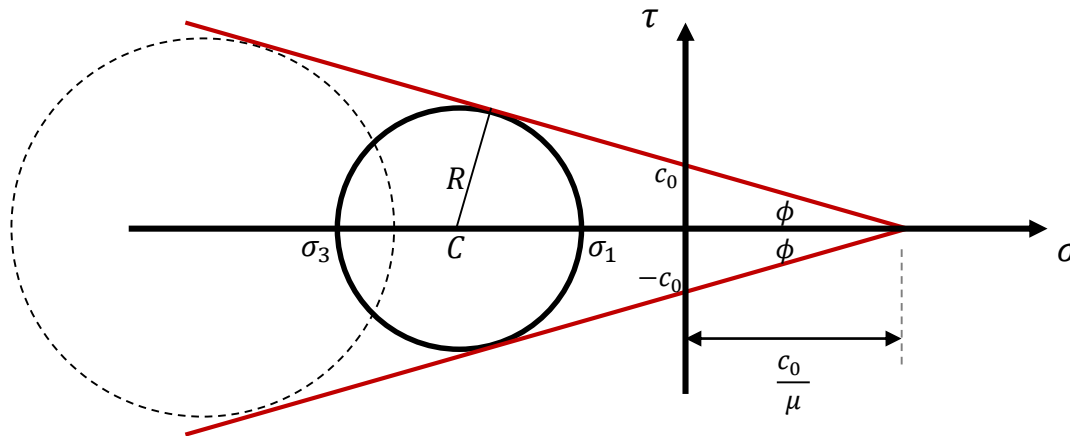
Mohr-Coulomb

$$R + \frac{k-1}{k+1}C - \frac{\sigma_c}{k+1} = 0$$

$$|\tau| - c_0 + \mu\sigma = 0$$

k and σ_c : material constants

R and C : are the radius and center of the Möhr circle in the plane $\sigma_1 - \sigma_3$



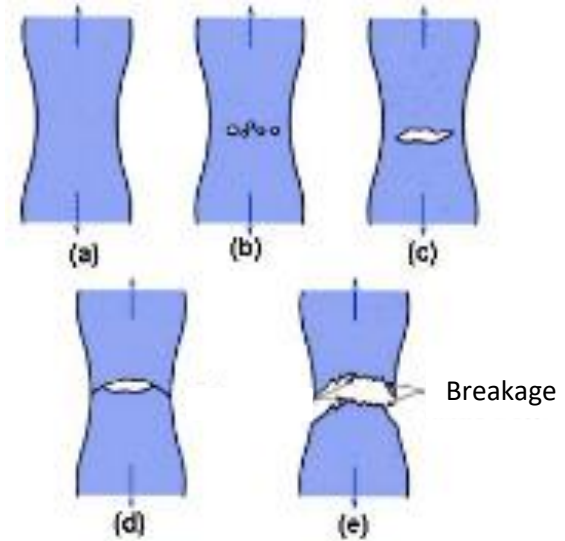
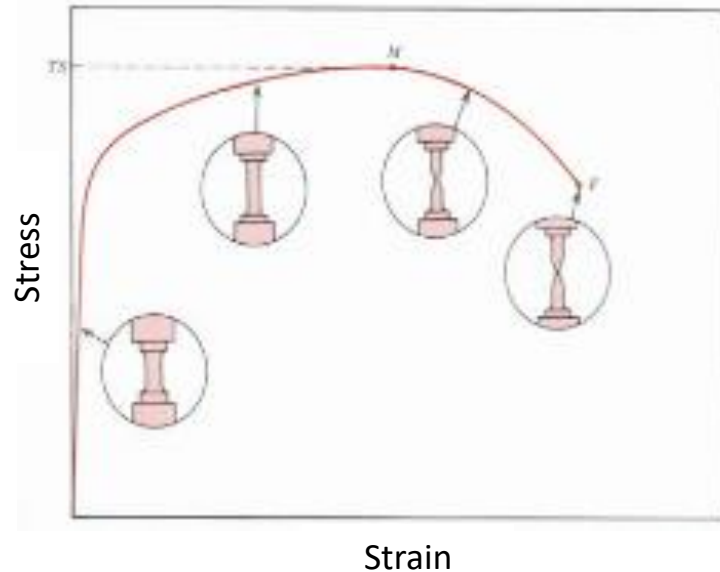
μ and c_0 are material properties (called coefficients of friction and cohesion, respectively).

In this case, the criterion has been represented in the "Mohr Circle" plane, instead of in the space of the principal stresses

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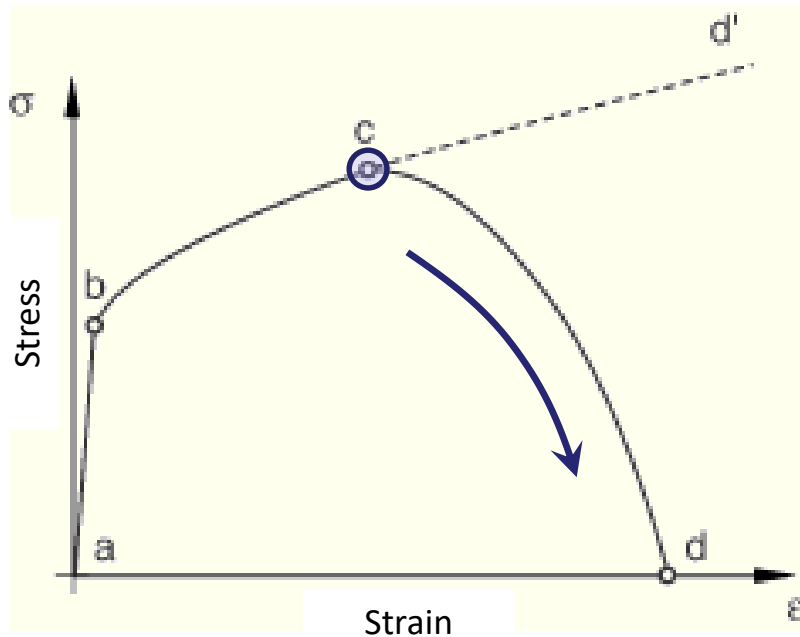
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Fundamentals



- a) Initial necking
- b) Formation of small cavities
- c) Crack appearance
- d) Crack propagation
- e) Final ductile breakage

Fundamentals

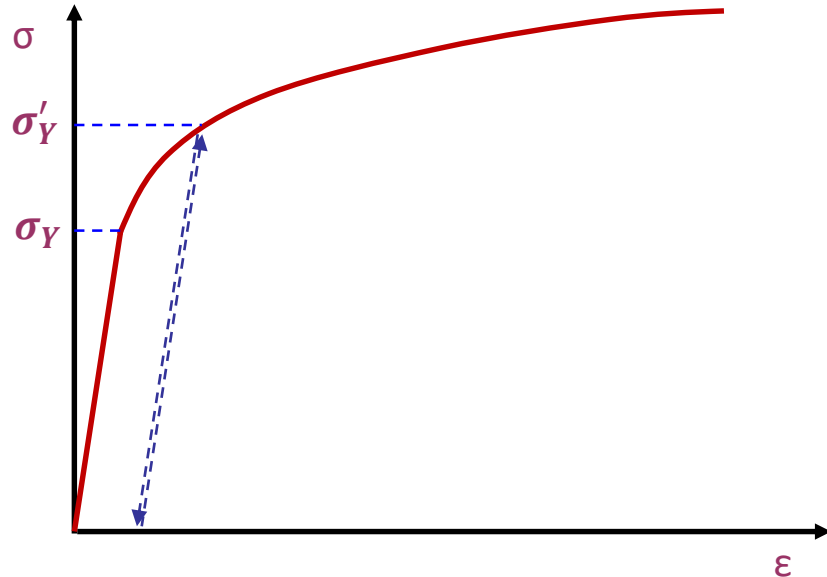


Complete loss of load capacity resulting from progressive degradation of material stiffness.

The stiffness degradation process is modelled using damage mechanics.

- **Section b-c:** apparition of plastic deformation.
Defined by **Strain Hardening Function**.
- **Point c:** onset of the reduction of the load capacity. Defined by **Damage Initiation Criteria**.
- **Section c-d:** stiffness degradation process.
Defined by **Damage Evolution Criteria**.

Strain Hardening



In a state of **uniaxial stress** several materials experimentally verify the following facts:

- The yield limit (or yield stress) of the material **increases with deformation**.
- This increase in the yield stress is called **strain hardening**.
- In an **unloading process** (in which there are no variations in plastic deformation) **the yield limit does not change** (no hardening).
- The increase of the yield limit is essentially related to a **variation of the plastic deformation** of any sign.

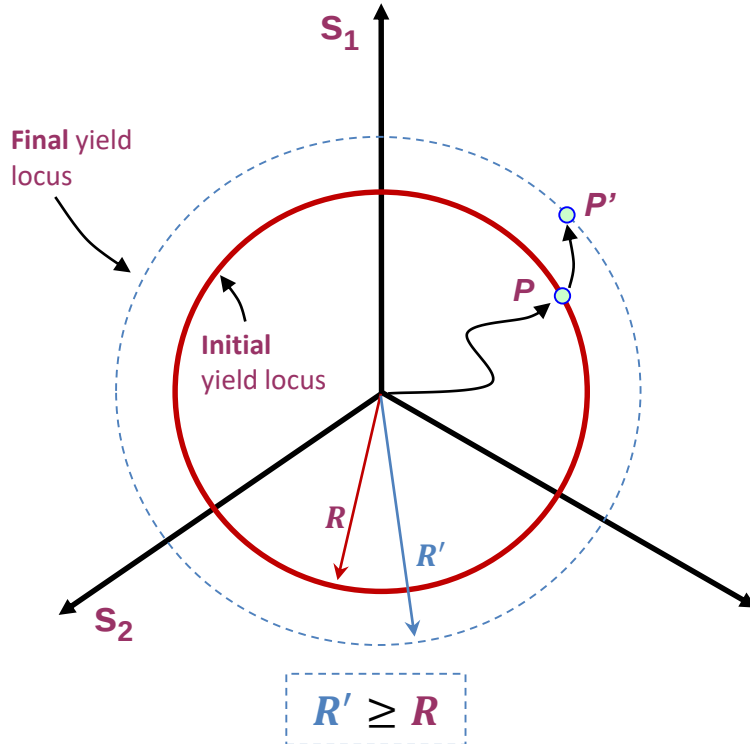
In a state of **multi-axial stress** (in which the concept of the yield limit is generalized by the plasticizing place concept), strain hardening takes place in those steps of the loading process in which, simultaneously:

- the yield criterion is fulfilled
- plastic strain varies

The yield limit (and therefore the yield surface) **depends on the whole history of plastic strains** of the material.

The yield limit, however, does **not vary in unloadings**.

Strain Hardening



If a material, whose **initial yield limit** satisfy the **Von Mises Criteria**, is capable of **hardening by plastic deformation**, then:

How does the **yield surface** vary during the hardening process?

If new plastic strain is imposed from P on the material and the material suffers strain hardening, this hardening results in an **expansion of the yield surface**.

This is what is known as **ISOTROPIC HARDENING**: the yield locus increases in size, without changing its shape, or its centre in the stress space

AN ANALYTICAL DESCRIPTION OF THE EVOLUTION OF THE YIELD SURFACE IS NECESSARY

Strain Hardening

Describe the SIZE CHANGE of the
YIELD SURFACE



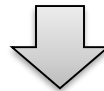
Describe the EVOLUTION of σ_Y during a
PLASTIC DEFORMATION process

Hardening Function

$$\sigma_Y = H(\bar{\varepsilon}^p)$$

The yield limit as a function of the
equivalent plastic deformation

$$\bar{\varepsilon}^p = \int d\bar{\varepsilon}^p \quad \text{where} \quad d\bar{\varepsilon}^p = +\sqrt{\frac{2}{3}} d\varepsilon_{ij}^p d\varepsilon_{ij}^p$$



Characterization Tests

Some hardening functions

$$\sigma_Y = H(\bar{\varepsilon}^p) = H\left(\int d\bar{\varepsilon}^p\right)$$

Cowper and Symonds

$$\sigma_Y = \sigma_0 \left[1 + \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right)^n \right]$$

- σ_0 : yield limit in uniaxial tensile test
- $\dot{\varepsilon}_0$: reference strain rate
- n : hardening exponent

Hollomon Law

$$\sigma_Y = K(\bar{\varepsilon}^p)^n$$

The initial yield limit (when plastic strain is 0) is null, so this law has not a well define initial yield limit

Lüdwick Law

$$\sigma_Y = A + K(\bar{\varepsilon}^p)^n$$

The initial yield limit in this case is:

$$\sigma_{Y0} = \sigma_Y(\bar{\varepsilon}^p = 0) = A$$

so this law has a well define initial yield limit

Some hardening functions

$$\sigma_Y = H(\bar{\varepsilon}^p) = H\left(\int d\bar{\varepsilon}^p\right)$$

Johnson-Cook Law

$$\sigma_Y(\bar{\varepsilon}_p, \dot{\bar{\varepsilon}}_p, T) = (A + B\bar{\varepsilon}_p^n) \left(1 + C \ln \frac{\dot{\bar{\varepsilon}}_p}{\dot{\bar{\varepsilon}}_0}\right) (1 - \Theta^m) \quad \text{where} \quad \Theta = \frac{T - T_r}{T_m - T_r}$$

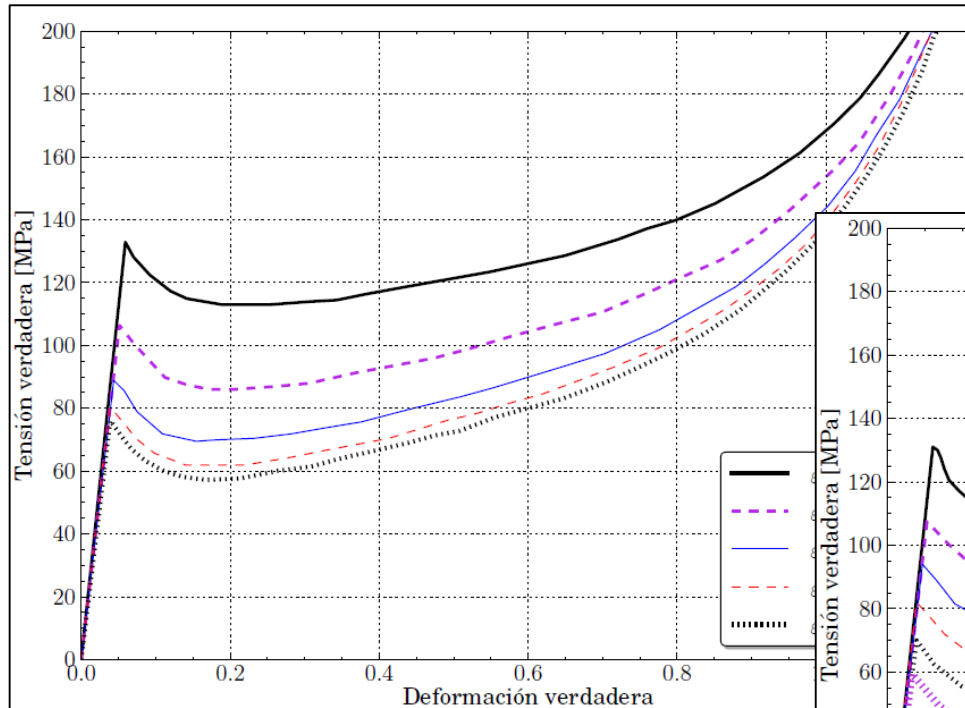
Strain rate hardening

Temperature softening

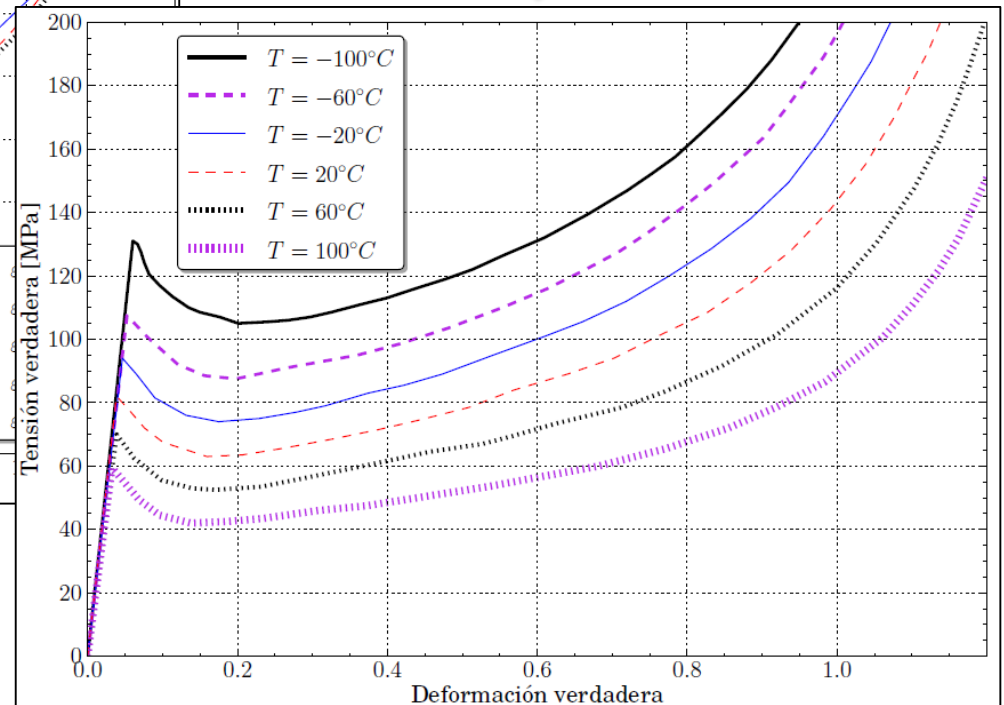
Some hardening functions

$$\sigma_Y = H(\bar{\varepsilon}^p) = H\left(\int d\bar{\varepsilon}^p\right)$$

Strain rate  hardening



Temperature  softening



Influencing Parameters

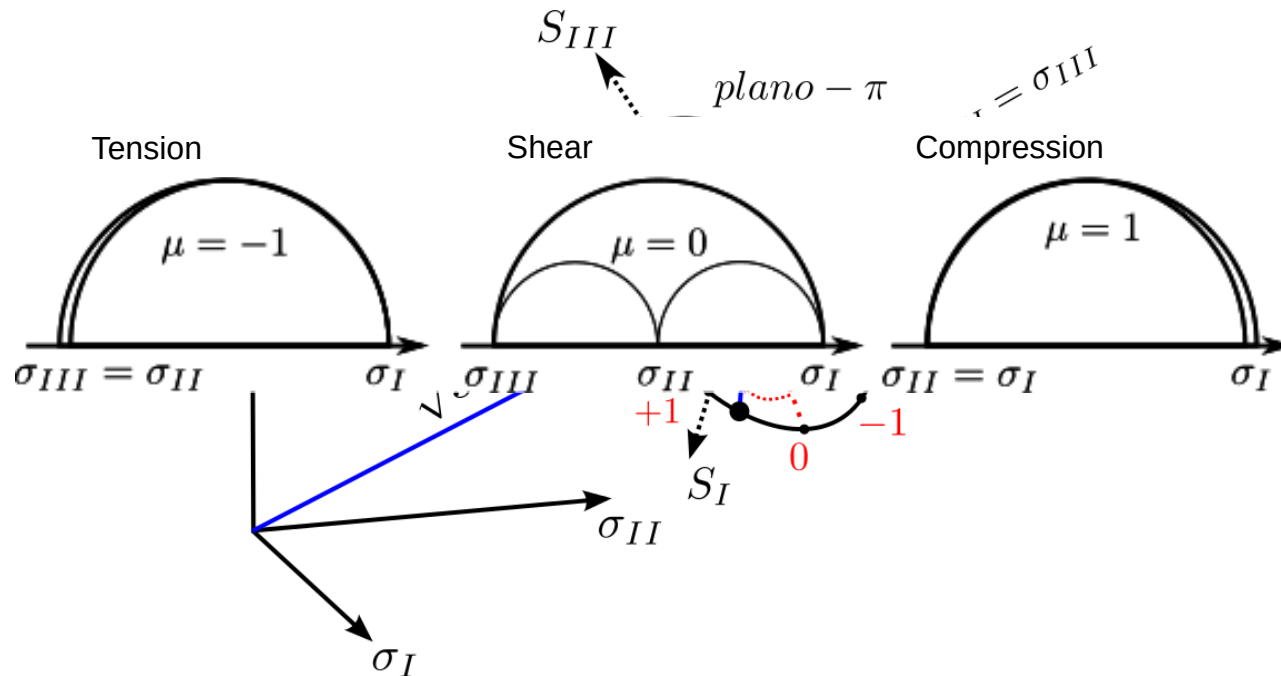
Influence of stress state on failure

Triaxiality

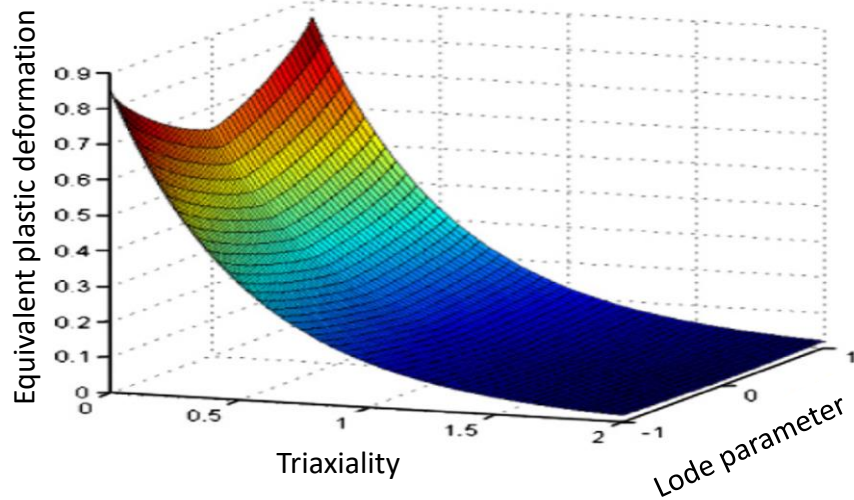
$$\eta = \frac{\sigma_m}{\bar{\sigma}}$$

Lode Parameter

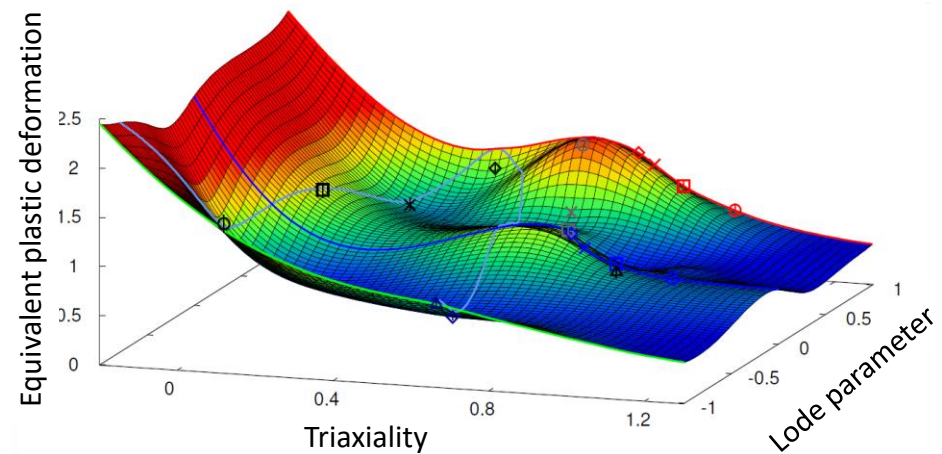
$$\mu = \frac{2\sigma_{II} - \sigma_I - \sigma_{III}}{\sigma_I - \sigma_{III}} \approx -\bar{\theta}$$



Influencing Parameters



1045 Steel



DP600 Steel

Complex study

Damage models classification

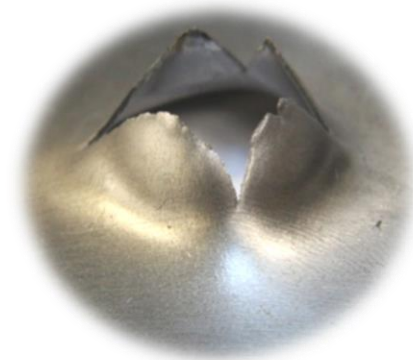
Models without degradation: in general, used for dynamic regime

Models with degradation: Can be used for both dynamic and quasi-static modes

Models without degradation

Tensile

Use **hydrostatic pressure** as a measure of failure for pressure tear or cut models.



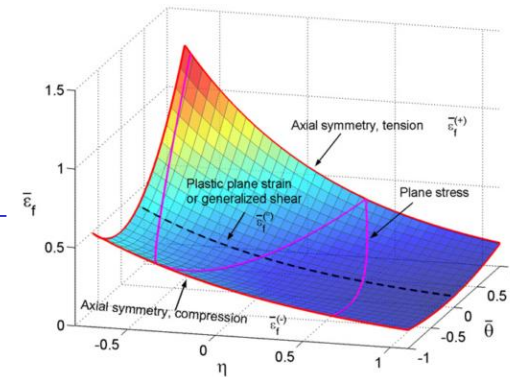
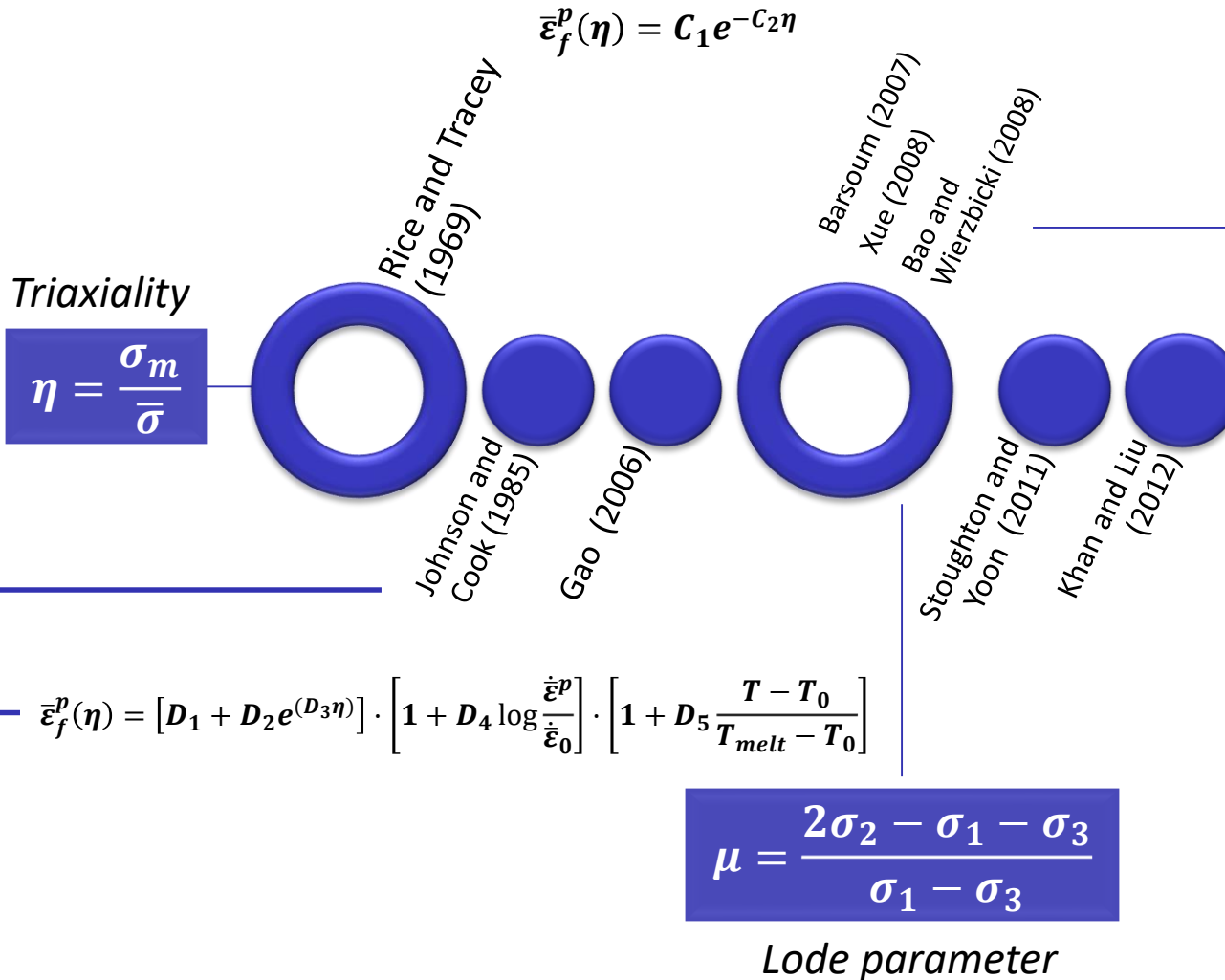
Shear

It is based on the **equivalent plastic strain**.
The failure occurs when the damage parameter equals 1.

$$\omega = \frac{\bar{\epsilon}_0^{pl} + \sum \Delta \bar{\epsilon}^{pl}}{\bar{\epsilon}_f^{pl}} = 1$$



Models with degradation



Models with degradation

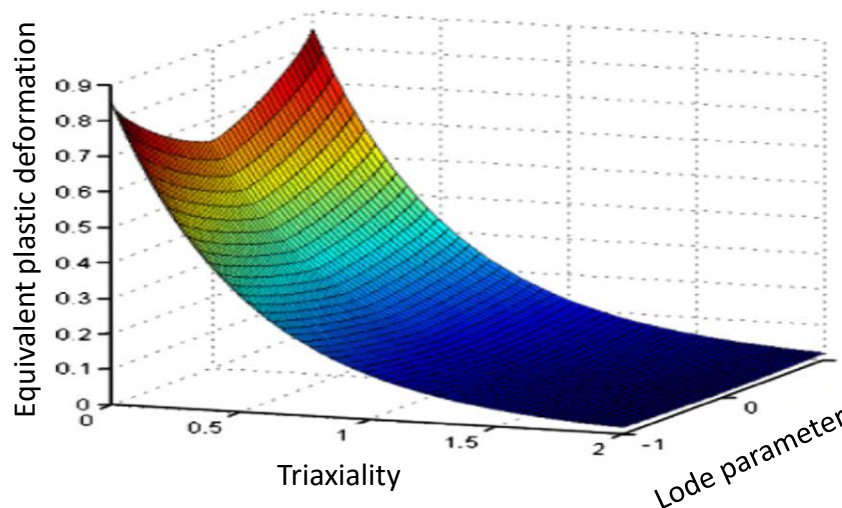
Rice and Tracey, 1969

Depends on **triaxiality** $\Rightarrow \bar{\epsilon}_f^p(\eta) = C_1 e^{-C_2 \eta}$

Johnson-Cook, 1985

Depends on **triaxiality, strain rate and temperature**

$$\bar{\epsilon}_f^p(\eta) = [D_1 + D_2 e^{(D_3 \eta)}] \cdot \left[1 + D_4 \log \frac{\dot{\bar{\epsilon}}^p}{\dot{\bar{\epsilon}}_0} \right] \cdot \left[1 + D_5 \frac{T - T_0}{T_{melt} - T_0} \right]$$

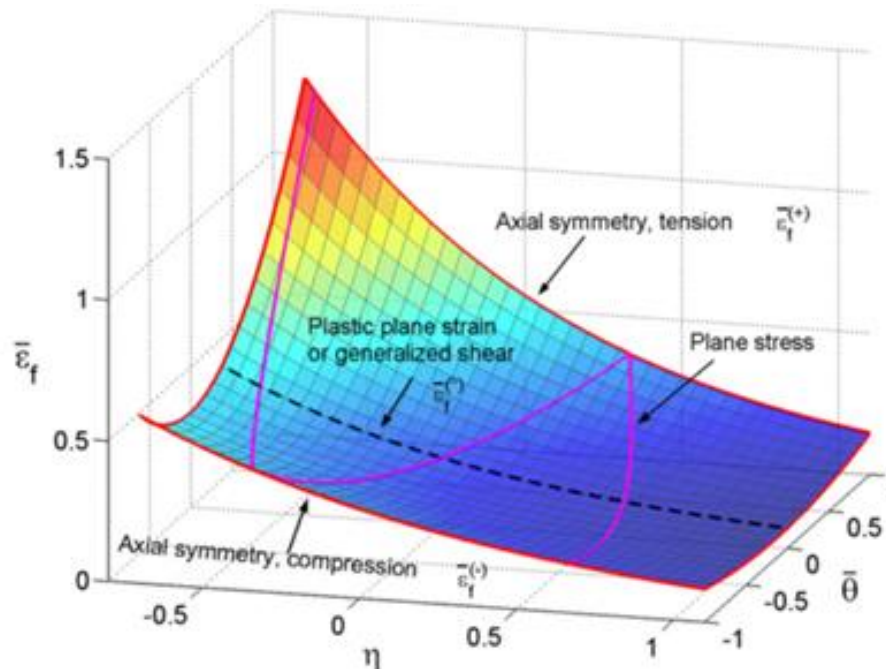


Models with degradation

Bao and Wierzbicki, 2008

Depends on **triaxiality** and **Lode parameter**

$$\bar{\varepsilon}_f(\eta, \bar{\theta}) = [D_1 e^{-D_2 \eta} - D_3 e^{-D_4 \eta}] \bar{\theta}^2 + D_3 e^{-D_4 \eta}$$



Models with degradation

Ductile Damage

Johnson-Cook Damage

Shear Failure

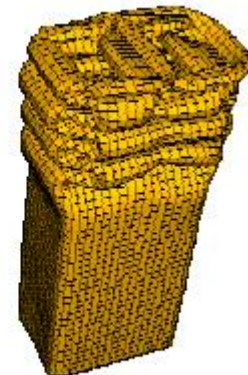
Bai-Wierzbizcki Model

Damage Initiation + Damage Evolution

Breakage does not appear if only **damage initiation** is defined.
It is necessary to consider also **damage evolution**.



Aluminum double-chamber
after dynamic impact



Analysis results with damage initiation
but no damage evolution

Models with degradation

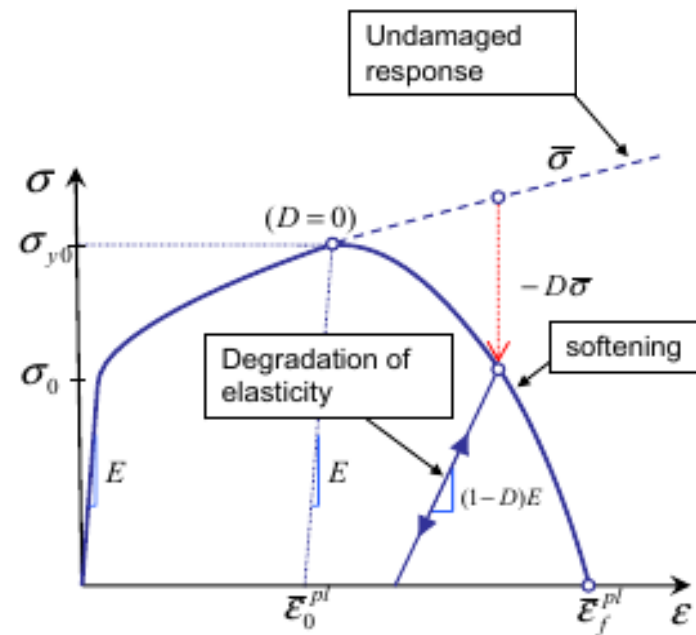
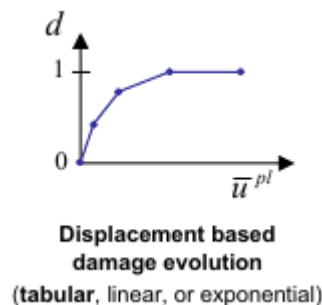
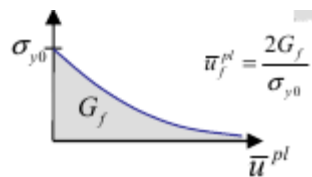
Damage Evolution

Defines the behavior of the material after the initiation of damage.

Degradation rate of the stiffness of the material once the initiation criterion has been satisfied.

The law of evolution of damage can be define by means of:

- Energy based damage evolution
- Displacement based damage evolution



Schematic representation of elastic-plastic material with progressive damage.

Models with degradation

Damage Evolution



Simulation **without**
damage evolution



Aluminum double-chamber
after dynamic impact



Simulation **with**
damage evolution

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Elastic material:

- **Stores** mechanical energy without dissipating it.
- If a **load** is applied **instantaneously**, the solid is **instantly deformed**.
- The stress-strain state **remains constant** until the load disappears.
- The tension state is of the "**restorer**" type: if the load ceases, the **shape recovers**.
- Stress σ depends on strain ε .

Viscous fluid:

- They are subjected to a non-hydrostatic stress state and **dissipate energy** without any possibility of storage.
- In a **shear stress state**, the fluid flows in a **stationary** manner.
- The stress state is **not of the "restorer" type**: if the stress ceases, the fluid particles do not return to their initial position.
- Stress σ depends on strain rate $\dot{\varepsilon}$.

Viscoelastic material:

- It can be considered to have an 'intermediate' behavior between elastic solid and viscous fluid.
- If a load is applied instantaneously, it undergoes instantaneous deformation, followed by another delayed deformation, increasing over time and which may or may not be limited.

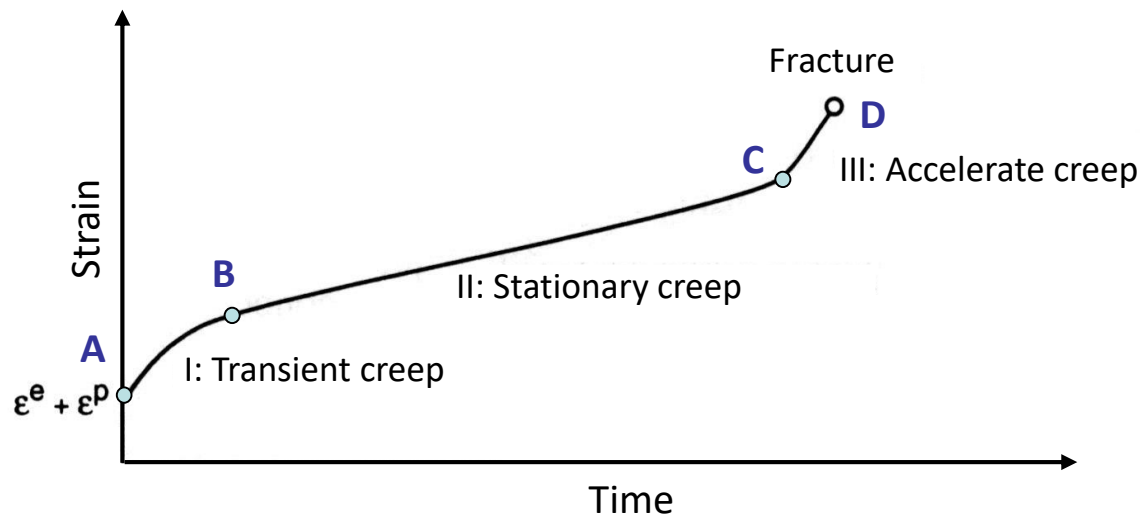
Just as the **elastic strain** is associated with the **displacement** of atoms from their equilibrium positions, the strain in the **visco-elastic case** is associated with the **diffusion** of atoms or molecules within the material.

Phenomenological aspects

A material can be recognized that as **VISCO-ELASTIC** (or visco-plastic) if the following phenomena are experimentally verified:

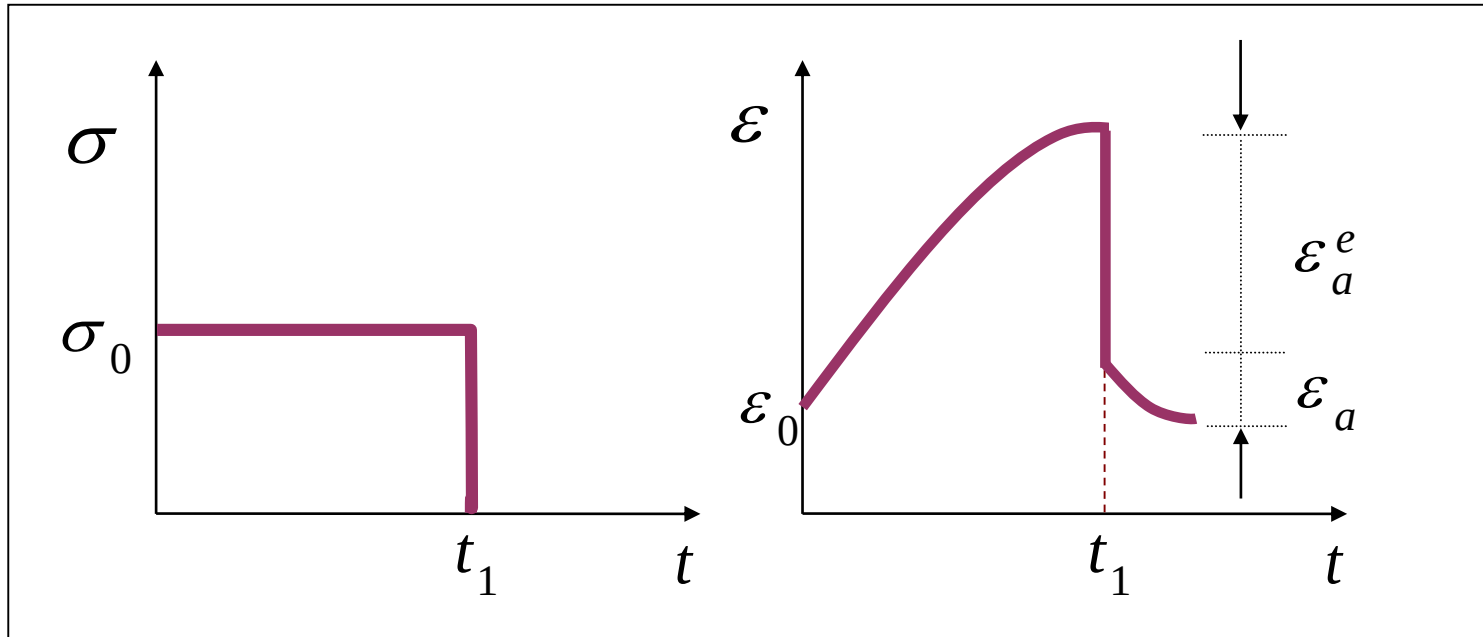
The **CREEP** phenomenon

The **RELAXATION** phenomenon



Viscoelastic Creep

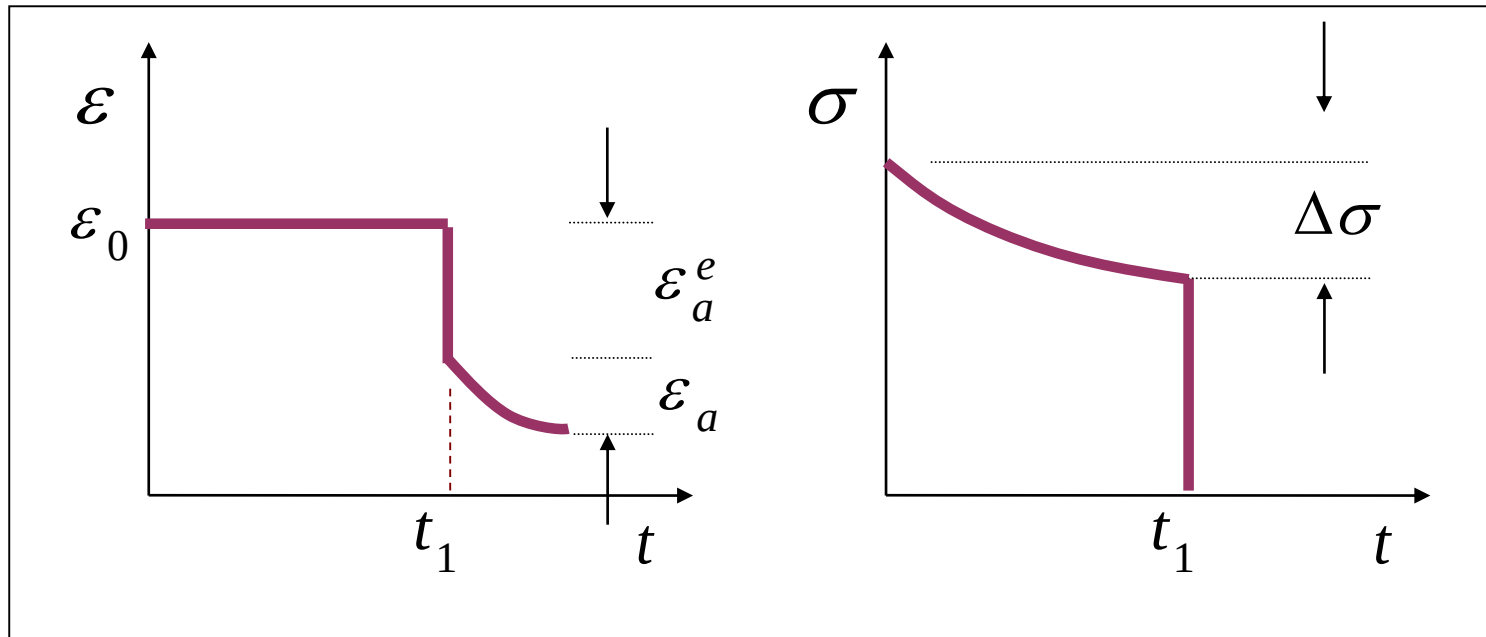
Creep Test



- If a stress σ_0 is applied instantly, which is then maintained constant over time, a gradual increase in the strain from the initial instantaneous value ϵ_0 is observed.
- If the specimen is unloaded (i.e. the stress disappears) in an instant t_1 , a drop in the strain (instantaneous drop at first and then gradual drop) is observed to a permanent value.

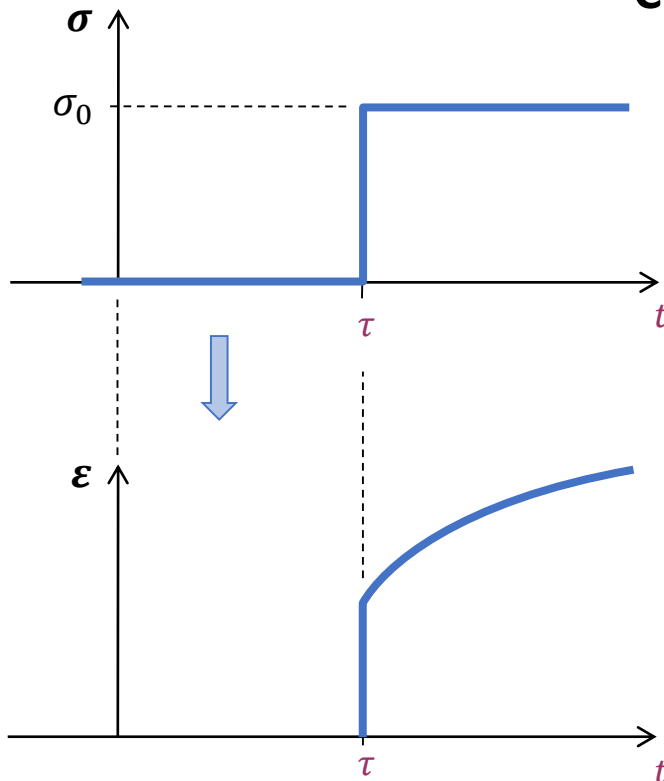
Stress Relaxation

Relaxation Test



- If the strain is maintained constant over time, a gradual decrease in stress is observed from an initial value that had been reached instantaneously.
- If the specimen is unloaded in an instant t_1 , a sudden and complete drop in tension is observed, as well as a drop in deformation (a sudden drop at first and then a gradual drop) to a permanent value.

Creep Function



A **constant stress** σ_0 is prescribed from the time τ :

$$\sigma = \sigma_0 H(t - \tau)$$

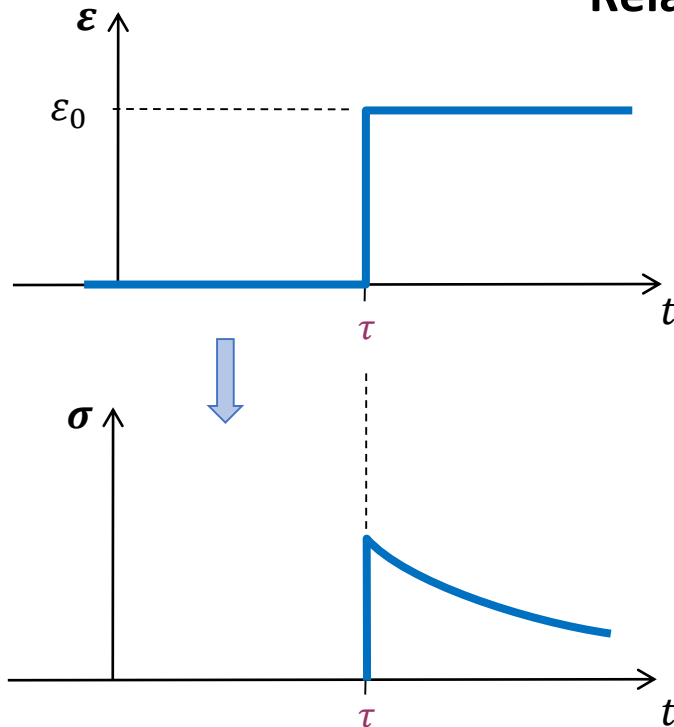
A **variable strain** is observed, described by the following function:

$$\epsilon = \sigma_0 J(t - \tau)$$

$J(t - \tau)$ is known as the **Creep Function**

The function $J(t - \tau)$ measures the strain response when the prescribed constant stress is unitary ($\sigma_0 = 1$)

Relaxation Modulus



A **constant strain** ε_0 is prescribed from the time τ :

$$\varepsilon = \varepsilon_0 H(t - \tau)$$

A **variable stress** is observed, described by the following function:

$$\sigma = \varepsilon_0 Y(t - \tau)$$

$Y(t - \tau)$ is known as the **Relaxation Modulus**

The function $Y(t - \tau)$ measures the stress response when the prescribed constant strain is unitary ($\varepsilon_0 = 1$)

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